Problem Chosen	2024	Team Control Number
A	MCM/ICM	2402168

Summary Sheet

Sex ratio change based on cellular automata model

Summary

The abundance or drastic decline of any species in the ecosystem can cause damage to the ecosystem, and sea lamprey, as a parasite that can alter the sex ratio, is very important for maintaining the stability of the ecosystem. The purpose of this study was to investigate the effects of changes in the sex ratio within the species of sea lampreys on themselves, their ecosystems, and the species within them.

First, we improve the fixed position selection and cell state of the traditional cellular automata algorithm, first establish a gender ratio change model on the two-dimensional plane, take the quadrilateral grid space of 100×100 as the cell space, define female, male, juvenile individuals as the cell state set, adopt the third-order extended Moor domain rule for the cell neighborhood, and select the periodic boundary condition to simulate infinite space. After 10,000 iterations of Matlab simulation, the proportion of males in lampreys changed from 56% to 78%. Based on the simulation results, it is concluded that in a larger ecosystem, the sea lamprey population can adjust the sex ratio according to changes in the environment, and its sex ratio has been in a dynamic process.

Second, we use the improved cellular automata model and the existing literature and data to know that the male and female sea lampreys will die after spawning, and if the larvae receive enough food supply in the juvenile period, the probability of becoming female in adulthood is greater, and the probability of becoming female is even less. Concluded that the lamprey population has the advantages of maintaining its own population, occupying a strong position in the competition with local primitive invaders, and the continuous evolution of the genome, and the conditions for obtaining resources cannot be guaranteed in the juvenile period.

Thirdly, we added three parasite populations on the basis of the Lotka-Volterra model, and at the same time considered the interspecific relationship between parasitism and competition, and reasonably set a series of parameters such as population capacity, birth rate, mortality rate and predation rate of each species, and the number of the three parasites increased significantly when the sex ratio of lamprey changed. According to the simulation results, it is concluded that when the change of sex ratio of lamprey has a stable impact on interspecific competition, the stability of the ecosystem decreases, and on the contrary, there is no significant impact on the stability of the ecosystem.

Finally, we used the Lotka-Volterra model of interspecific relationships, considering that lampreys have no natural predators and can be parasitized by parasites, and conclude that changes in the sex ratio of lamprey populations can provide advantages to other species in the ecosystem (e.g. parasites).

Keywords: Cellular automata model Lotka-Volterra model Changes in sex ratio

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1 Introduction

1.1 Problem Background

Many species exhibit a 1:1 sex ratio at birth, but others deviate from the uniform sex ratio, a phenomenon known as adaptive sex ratio change, and Lampreys are one such species.Lampreys are parasites that latch on to other species for nutrients and have few predators.After laying eggs, both parents die and only the larvae are left, and the growth rate of the larvae is affected by the food supply, and the growth rate of the larvae determines the sex of the sea lampreys, so the sex ratio of the sea lampreys will change with the external environment, that is, the food supply.In environments where food is plentiful, the larvae grow faster, with male lampreys accounting for about 56% of the total population.In a food-deprived environment, young lampreys eat less and grow slowly, and become males when they reach adulthood, resulting in up to 78% male lampreys.

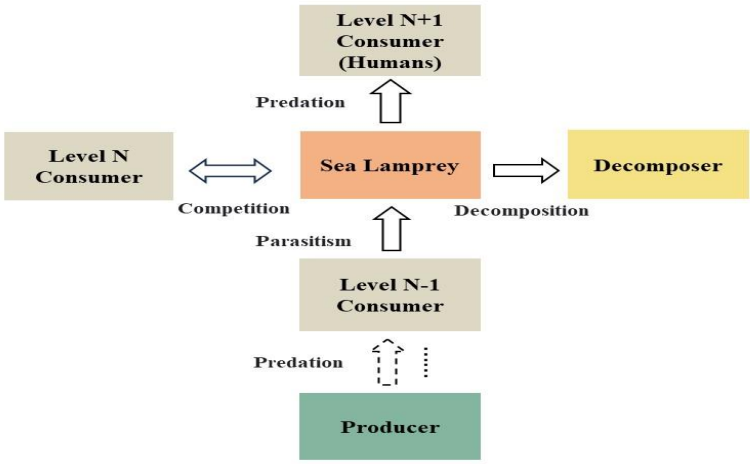

Figure1 Lampreys food structure map

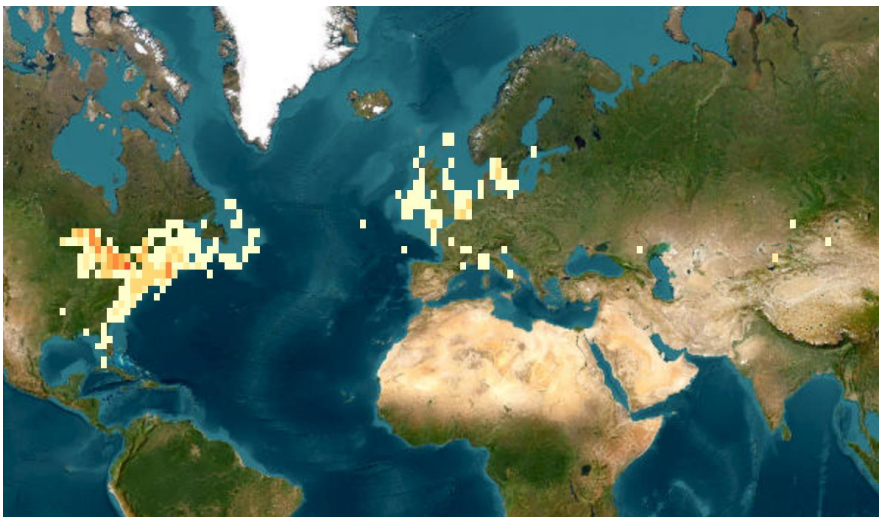

Figure2 **Lampreys distribution map**

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

Problem 1，When a population of lampreys can adjust its sex ratio on its own in response to changes in the environment, what are the implications for the larger ecosystem in which it resides.

Problem 2，As a species that has been around for 360 million years, lampreys have their own unique strengths and weaknesses.

Problem 3，Lampreys, as consumers in the ecosystem, have implications for the stability of the ecosystem when their sex ratio changes.

Problem 4，One or more of the four relationships of predation, competition, parasitism, and mutualism among different species, whether changes in the sex ratio of lampreys have an auxiliary effect on other species

in the ecosystem.

1.3 Our Work

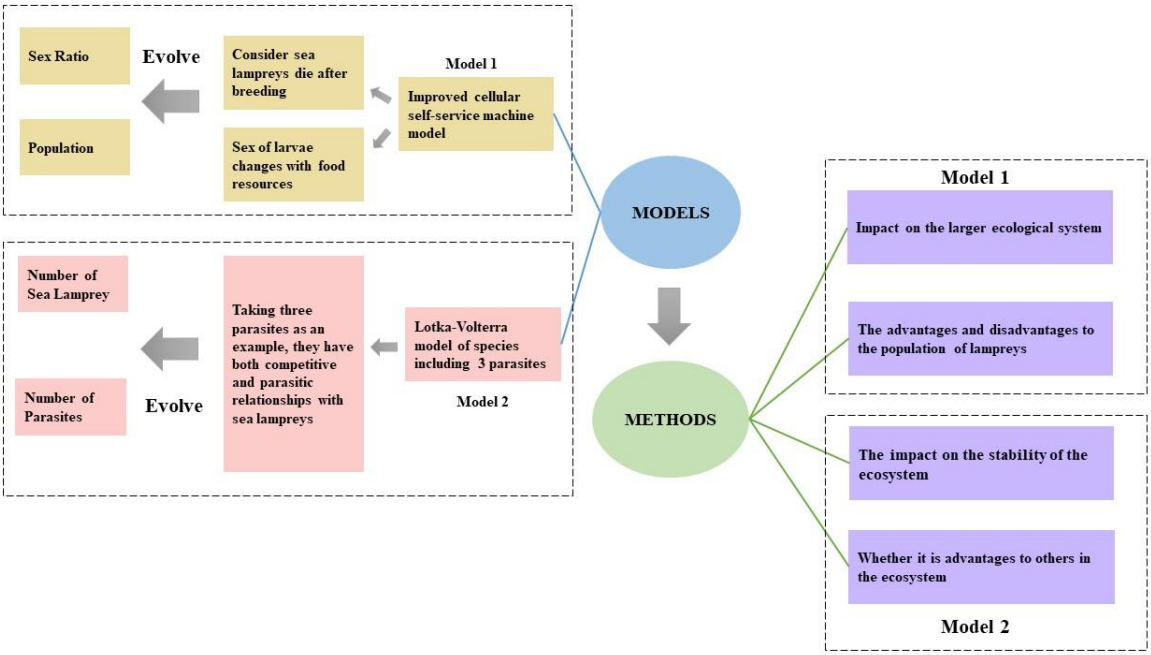

Figure3 Paper flow chart

2 Assumptions and Justifications

1.Lampreys initially live in well-fed conditions, i.e. males make up 56%.

Since the ecosystem has a certain self-regulation ability, almost all species are in a state of stable change and normal life in the absence of major human interference and natural disasters, which is consistent with the reality.

2.Suppose there are three parasites in the same ecosystem as lampreys, and these three parasites have both parasitic and competitive relationships with lampreys.

In nature, there are already complex food chain and food web relationships among various species. In order to simplify the calculation, only three kinds of parasites are selected in this paper to simulate the relationship with lampreys population. Although there are only three kinds, the relationship between parasitism and competition is also included.

3.It is assumed that the population size of lampreys is higher than that of other parasites

Lampreys have survived 360 million years on Earth with few predators, so there must be something unique about them that has kept them from being eliminated by natural selection, and it makes sense that their population size is higher than other parasites.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
C	Cellular space
S	Set of cell states
V	Cellular neighborhood
f	Cellular state transition rule
N_f	Population number of female sea lamprey
N_m	Population number of male sea lamprey
$E1$	Predation rate of lampreys by parasites
K_{p1}	Population capacity of the parasite
$P1$	The number of parasites

4 Model 1: Sex ratio change model of lampreys

4.1 Establishment of sex ratio change model

As a discrete dynamic system in time, space and state, cellular automaton (CA) was first proposed by V on Neumann, which is mainly composed of four parts, including cell, cellular space, cellular neighbor and evolution rules[1]. It can simulate complex evolution process through limited cell states and specific evolution rules.

In this paper, a two-dimensional cellular automaton model was used to simulate the proportion of male lampreys under different conditions, and the cellular automaton model was expressed as $CA=(C,S,V,f)$, In this expression, CA represents the cellular automata model, C represents the cellular space, S represents the set of cell states, V represents the cellular neighborhood, f represents the cellular state transition rule.

(1)Cellular space C: Let C be a quadrilateral grid space of 100*100, representing the living area of lampreys. A number of central points were randomly selected to construct a 3*3 space, representing cells, and a certain proportion of lampreys were allocated to each cell according to its space size. Lampreys have complex mating and reproduction within the cell space and between adjacent cells. Cellular $C=\{(x_C,y_C)\in C\}$ represents its central point coordinate value, then the cellular space can be expressed as

$$C=\{(x_c,y_c)\in 100^2 \mid 1\leqslant x_c\leqslant 100, 1\leqslant y_c\leqslant 100\} \tag{1}$$

(2)Set of cell states S: According to the data, there are three gender states of lampreys: female, male and undetermined individual, in which 0 is used to represent female, 1 is used to represent undetermined individual, and 2 is used to represent male. Then the set of cell states S can be expressed as

$$S=\{0,1,2\} \tag{2}$$

(3)Cell neighborhood V: In the cellular neighborhood, the 3-order Moor domain rules are extended, and periodic boundary conditions are selected to simulate the infinite space.

(4)Cellular state transition rule f:

If a cell is an undetermined sex individual, the sex is determined by the male proportion in its cell neighborhood V. When the male proportion is greater than 78%, the undetermined sex individual will become female, and vice versa.

If a cell is a female individual, and if there are males in the cell neighborhood V, then it produces offspring with one of the male individuals, at which time the female and the male it copulates with die and is replaced by an undetermined individual.

4.2 Solving the sex ratio change model

According to the question, when food is sufficient, the proportion of

males accounts for about 56% of the total number of lampreys. Therefore, this paper assumes that in an area with sufficient food, there are 10,000 lampreys, of which 1000 are still in the undetermined stage of growth and development, and that of the remaining 9000 lampreys, 56% (5040 males) and 44% (3960 females) will be found. Using Matlab programming, after 10,000 iterations, it can be obtained that the proportion of male lampreys becomes 75.46%, which is very close to the proportion of male lampreys at 78% under the condition of insufficient food supply.

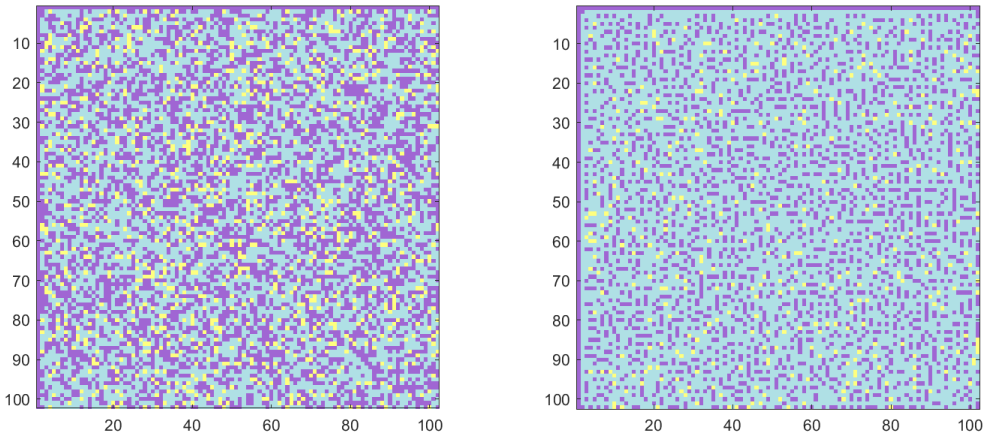

Figure 4: Schematic diagram before (left) after (right) iteration of cellular automaton

Among them, purple represents the female individual, yellow represents the undetermined sex individual, and blue represents the male individual. It can be clearly seen from Figure 4 that the proportion of male individuals has increased from the original 56% to 75.46%.

4.3 interpretation of result

4.3.1 Effects of sex change in lampreys on ecosystem

Natural resources and space are always limited. It is assumed in this paper that the sex ratio of lampreys is relatively normal at the beginning, with the male lampreys accounting for about 56%, but with the increase of population density, the decrease of food and space resources, and the intensifies intraspecies competition, the birth rate of lampreys population decreases, the mortality rate increases, and the population number declines. The slow dominance of the strong male lampreys led to an imbalance in the sex ratio, which reached about 78% males, so the lampreys population changed its sex ratio on its own.

Considering only the long time period from 56% to 78% of lampreys males, the number of competitive lampreys populations will eventually decline, and the number of parasitic lampreys populations will also decline, and in the larger ecosystem, fewer consumers will lead to more producers and non-living matter and energy, more decompers, and increased ecosystem stability.

4.3.2 Advantages and disadvantages of lampreys population

advantage:

(1) Maintain their own population: Lampreys can adapt to different resource environments by changing their sex, which can effectively improve

the pregnancy rate and increase the population number in the living space with abundant food.

(2) It occupies a strong position in competition with local original invaders: in the Great Lakes region, the sexually changeable lampreys can parasitize the protoinvader lake trout, reducing its population[2], indirectly reducing the number of other local aquatic organisms preyed on by the protoinvader, protecting local aquatic organisms to a certain extent, and thus providing them with more food resources in the competition.

(3) Continuous evolution of the genome: Numerous medical studies[3] have shown that lampreys, as gnathen vertebrates, have RNA related to immune and evolutionary mechanisms in their genomes, which plays a significant role in their own racial adaptability.

disadvantage:

Access to resources during adolescence cannot be guaranteed: sea lampreys are egg-laying organisms in anadroits, and female sea lampreys need to migrate from the original living area to the river to lay eggs[4]. Their parents die after laying eggs, so it is impossible to guarantee whether the local food resources will change after the birth of the larvae, and the larvae feed on plankton and debris, and cannot prey on parasites before the juvenile transformation.

5 Model 2: Interspecific Relationship Model

5.1 Establishment of interspecific relationship model

The sea lamprey, within its habitat, undergoes long-term coevolutionary interactions with other organisms, giving rise to interspecific relationships depicted in Figure 5. In this figure, arrows denote the direction towards predators or parasites.

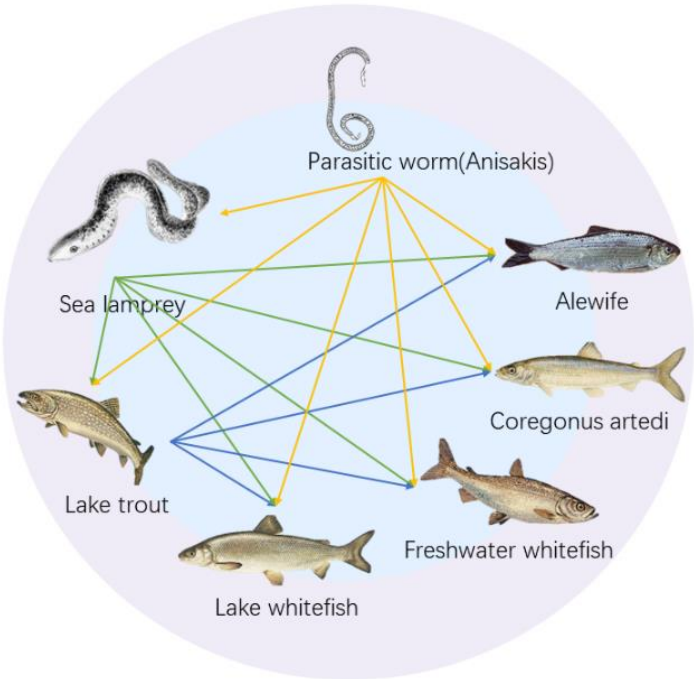

Figure 5 Interspecific relationships in ecosystem (sections)

Fish species coexisting in a particular ecosystem engage in competitive relationships due to the shared resource space. Additionally, the sea lamprey, along with other parasites such as the Anisakis, competes as a parasite. Simultaneously, certain parasites can establish a parasitic relationship by infesting the lamprey.

In ecology, the mutual adaptation and mutual promotion between predators and prey give rise to a unique relationship, exemplified by the Lotka-Volterra model, which stands as a classical dynamical model between the two. A.J. Lotka and V. Volterra independently formulated models for predator-prey systems and competitive systems, and E.P. Odum later extended these models to mutualistic systems, collectively categorizing these three models as the Lotka-Volterra model[5]. The Lotka-Volterra model, designed for the coevolution of two populations, is employed to describe the characteristic changes in quantity between predators and prey. It has found widespread application in the study of population growth models in interactions between two species[6].

This study postulates the coexistence of three parasitic species with the sea lamprey in the same ecological system. These three parasites not only compete with the sea lamprey for food resources, but their population numbers are also influenced by the availability of resources and the population size of the sea lamprey. Simultaneously, these three parasites engage in a parasitic relationship with the lamprey, establishing both parasitic and competitive interspecific interactions. Furthermore, parameters such as birth rates, death rates, and growth rates of the sea lamprey need to be considered. To depict higher-order interspecific interactions (HOIs), a Lotka-Volterra model is to be constructed.

The population growth rates of the sea lamprey (for both males and females) are represented by the following equations:

$$\frac{dN_f}{dt}=\left(r_f\times N_f\times\left(1-\frac{N_f}{K_f}\right)-S_f\times N_f\right)\times(1-P_f) \tag{3}$$

$$\frac{dN_m}{dt}=\left(r_m\times N_m\times\left(1-\frac{N_m}{K_m}\right)-S_m\times N_m\right)\times(1-P_m) \tag{4}$$

The population growth rates of the three parasitic species are represented by the following equations:

$$\frac{dP1}{dt}=\left(r_{p1}\times P1\times\left(1-\frac{P1}{K_{p1}}\right)-S_{p1}\times P1\right)\times(1-E1) \tag{5}$$

$$\frac{dP2}{dt}=\left(r_{p2}\times P2\times\left(1-\frac{P2}{K_{p2}}\right)-S_{p2}\times P2\right)\times(1-E2) \tag{6}$$

$$\frac{dP3}{dt}=\left(r_{p3}\times P3\times\left(1-\frac{P3}{K_{p3}}\right)-S_{p3}\times P3\right)\times(1-E3) \tag{7}$$

The predation rates of the sea lamprey on the three parasitic species are represented by the following equations:

$$E1=k1\times N_f\times P1 \tag{8}$$

$$E2=k2\times N_f\times P2 \tag{9}$$

$$E3=k3\times N_f\times P3 \tag{10}$$

In these equations, N_f and N_m represent the population numbers of female and male sea lampreys, r_f and r_m denote the birth rates of female and male sea lampreys, K_f and K_m represent the population capacity of female and male sea lampreys, respectively, S_f and S_m signify the death rates

of female and male sea lampreys. Similarly, $P1$, $P2$ and $P3$ are the population numbers of Parasite 1, Parasite 2, and Parasite 3, r_{p1}, r_{p2} and r_{p3} denote the birth rates of Parasite 1, Parasite 2, and Parasite 3, K_{p1}, K_{p2} represent the population capacity of Parasite 1, Parasite 2, and Parasite 3, S_{p1}, S_{p2} and S_{p3} signify the mortality of Parasite 1, Parasite 2, and Parasite 3, $E1$, $E2$ and $E3$ are the predation rates of sea lampreys by three parasites, $k1$, $k2$ and $k3$ are parameters related to predation rate.

5.2 Solving the interspecific relationship model

Due to the lack of natural predators for the sea lamprey, it is assumed that the carrying capacity (K-value) of the sea lamprey population number is higher than that of other parasites. Additionally, both male and female individuals of the sea lamprey are initially more numerous than those of other parasites. Various parameters, including population capacity, birth rate, death rate, predation rate, and other factors for each species, have been configured as shown in Table 2.

Table 2 Lotka-Volterra model parameters

The name of the parameter	Parameter value
Birth rate of female sea lampreys	0.05
Birth rate of male sea lampreys	0.05
Population capacity of female sea lampreys	1000
Population capacity of male sea lampreys	1000
Mortality of female sea lampreys	0.03
Mortality of male sea lampreys	0.03
Predation rate parameter of parasite 1	0.015
Predation rate parameter of parasite 2	0.02
Predation rate parameter of parasite 3	0.025

Continued:

The name of the parameter	Parameter value
Birth rate of parasite 1	0.04
Birth rate of parasite 2	0.035
Birth rate of parasite 3	0.03
Population capacity of parasite 1	500
Population capacity of parasite 2	400
Population capacity of parasite 3	350
Mortality of parasite 1	0.02
Mortality of parasite 2	0.025
Mortality of parasite 3	0.03
Initial infection rate of parasite 1	0.2
Initial infection rate of parasite 2	0.25
Initial infection rate of parasite 3	0.3
Initial food availability	0.6

The Lotka-Volterra model was established and solved by Matlab, and the number of other three parasites changed when the sex ratio of lamprey changed from 56% to 78%, as shown in Figure 6 and Figure 7 below.

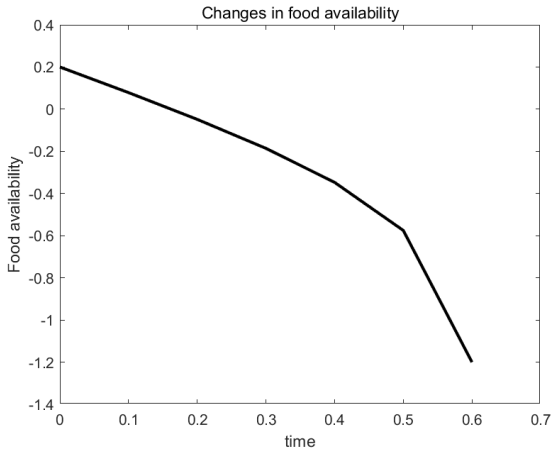

Figure 6 food availability changes

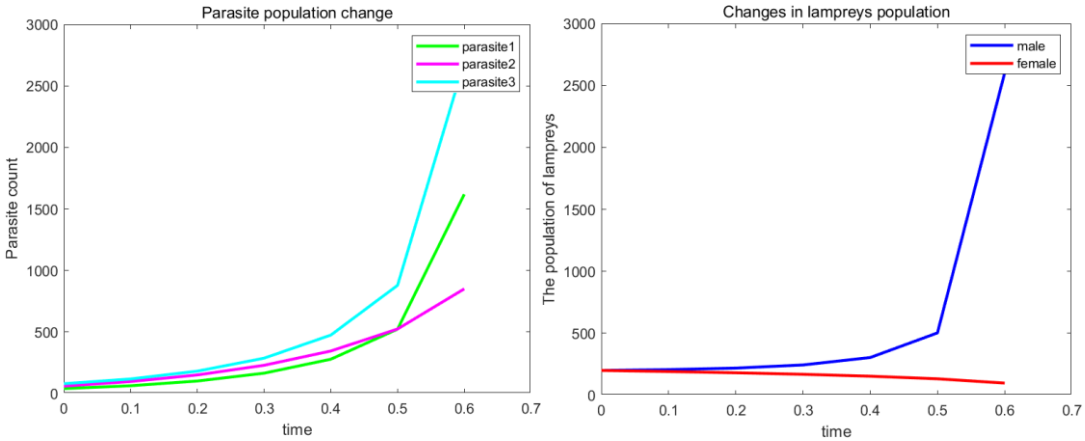

Figure 7 in the population numbers of three parasites Changes (left)

Male and female population of sea lampreys changes (right)

It is evident from the figure that as the number of male sea lampreys increases, the number of female lampreys decreases, while the number of the three parasites also increases significantly, and the available food decreases dramatically.

5.3 Analysis of results

5.3.1 Effect of sex ratio of sea lamprey on ecosystem stability

1.Ecosystem stability refers to the ability of an ecosystem to maintain its structure and function in the face of internal and external changes. Ecosystem stability involves many different aspects:

(1) Species diversity: the number and abundance of various different species present in the ecosystem.

(2)Intensity of interspecies interactions: the strength of interactions between different species in ecosystems.

(3) Energy flow and material cycle: the flow and circulation of energy

and key nutrients in the ecosystem.

(4) Ecosystem resilience: The ability of an ecosystem to maintain its stability in the face of disturbances.

(5) Ecosystem resistance: the stability and resilience of ecosystems in the face of disturbances.

2. Analysis of sex ratio change in lamprey:

(1) When the sex ratio of sea lampreys tends to be 56%:44%, the birth rate of this population increases, but because the parents of sea lampreys die after breeding, and it takes 3-5 years for sea lamprey larvae to reach the adult state that can parasitize other organisms, the population of other species increases in the short term, and the interspecific competition becomes stronger at this stage, and the stability of the ecosystem is damaged to a certain extent[7], while the population increases after the ecosystem moves into new sea lampreys, because sea lampreys are in a predatory position in the ecosystem[8] At this stage, the population number of other organisms decreased, but the species diversity remained unchanged, the interspecific competition of these organisms became weaker, and the stability of the ecosystem was restored to a certain extent.

(2) When the sex ratio of sea lampreys tends to be 22%:78%, the birth rate of this species decreased, but due to the small number of sea lampreys that died after breeding, there was little difference between them before the

breeding period, the species diversity remained unchanged, the interspecific competition of other organisms was stable, and the stability of the ecosystem remained in a certain balance.

3. Conclusion

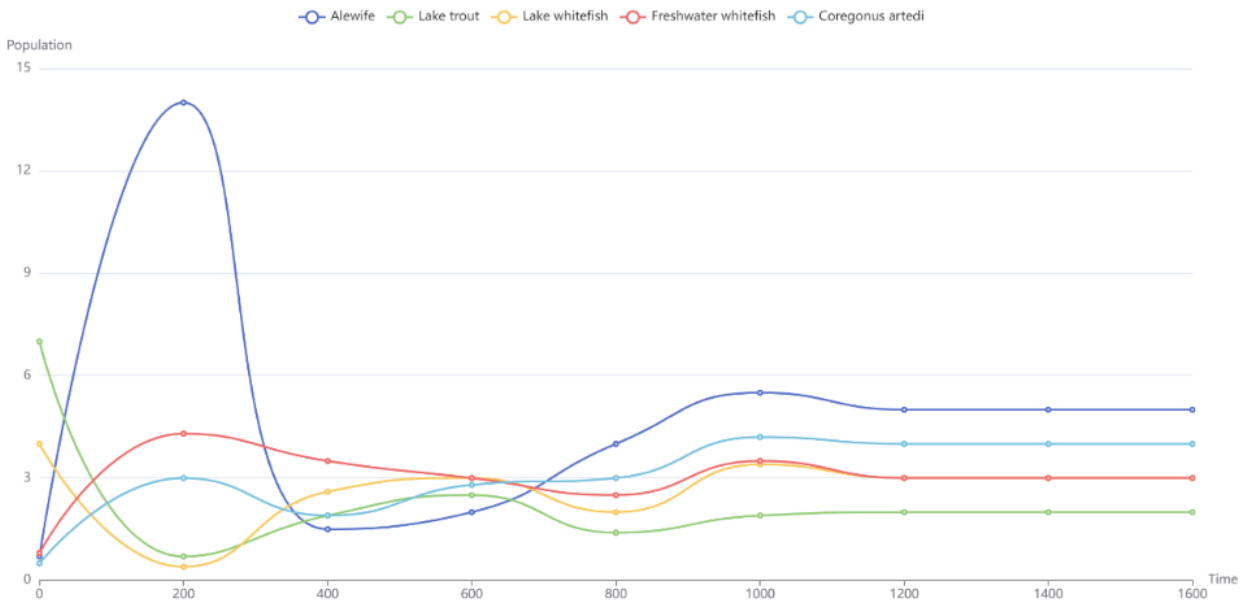

Figure 8 Simulation of population dynamics over time

When the change of sex ratio of sea lamprey has a stable impact on interspecific competition, interspecific competition has a negative impact on the average total biomass, reduces population synchronization, increases the variability of total biomass, and reduces the stability of the ecosystem.

When the change of sex ratio of sea lamprey had an unstable effect on interspecific competition, interspecific competition had no significant effect on the average total biomass, and had no significant effect on the stability of the ecosystem.

5.3.2 Effect of sex ratio in sea lamprey on other species

The resuts showed that the population number of the other three parasites increased gradually in the short term when the food resources were gradually decreasing. This is due to the fact that there is no restriction of natural predators for them, the population number of sea lampreys will increase when food resources are sufficient, and the population of three parasites hosting organisms, including sea lampreys, will gradually increase, which also indicates that parasitism on sea lampreys can offset the negative feedback brought by competing hosts with sea lampreys to a certain extent. After a long period of development, due to the decline in food availability, the population of most species in the ecosystem gradually decreases, and the population number of lampreys and parasites also decreases, eventually reaching a state of steady fluctuation. Thus, ecosystems with changing sex ratios in sea lampray population number can provide advantages to other species in the ecosystem, such as parasites.

6 Model Evaluation and Further Discussion

6.1 Strengths

1.In this paper, a two-dimensional cellular automaton model is used to simulate the changing process of the sex ratio of lampreys. The results verify that in nature, the ecosystem is dynamic and has a certain self-regulation ability, which is consistent with the facts, proving that the research in this

paper is scientifically based and credible.

2.In this paper, when using Lotka-Volterra model to study the interspecies relationship, three parasites that can parasitize Lampreys were selected. In this way, there are both parasitic and competitive relations between parasites and lampreys, which simplifies the calculation and is largely consistent with the situation in nature, which is an ingenious hypothesis.

3.The Lotka-Volterra model is one of the classical dynamic models between predator and prey, and this model can be used to restore the real situation.

6.2 Weaknesses

When studying the impact of the change of sex ratio of lampreys on other species and the larger ecosystem, this paper only considers the competition and parasitic relationship between lampreys and other species, and the remaining two inter-species relationships and intraspecific competition of lampreys are not involved. Moreover, the constructed ecosystem is too small, with only four species including lampreys.

6.3 Further Discussion

Model improvements: When studying the effects of changes in the sex ratio of lampreys on other species and the larger ecosystem, the ecosystem needs to be large enough to simulate all four species relationships and intra-

specific competition in lampreys. In addition, the study period can be extended to look at the long-term changes in the sex ratio of lampreys and their effects.

7 Conclusion

1.Lamprey populations can adjust their sex ratio according to environmental changes, and their sex ratio has been in a dynamic process of change.

2.Lampreys have the advantages of maintaining their own population, having a strong position in competition with the local original invaders, continuous evolution of the genome, and the disadvantages of obtaining resources in the juvenile period.

3. When the change of sex ratio of sea lampreys had a stable effect on the interspecific competition, the interspecific competition had a negative effect on the average total biomass, decreased the population synchronization, increased the variability of the total biomass, and decreased the stability of the ecosystem; otherwise, it had no significant effect on the ecosystem stability.

4. Ecosystems with changing sex ratios in lamprey populations can provide advantages to other species in the ecosystem, such as parasites.

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